

Mathematics Reference: Simplification

Educational Series

December 29, 2025

1 The Order of Operations: VBODMAS

To solve any mathematical expression correctly, follow the ****VBODMAS**** rule in sequence:

A V (Vinculum/Bar): Operations under a bar $\overline{x - y}$ are solved first.

B B (Brackets): Solve in order: Small () $\rightarrow$ Curly { } $\rightarrow$ Big [].

C O (Of): Equivalent to multiplication but solved before division (e.g., 50% of 100).

D D (Division): $\div$

E M (Multiplication): $\times$

F A (Addition): $+$

G S (Subtraction): $-$

2 Essential Algebraic Identities

Many simplification problems are based on these identities:

- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a - b)^2 = a^2 - 2ab + b^2$
- $(a^2 - b^2) = (a + b)(a - b)$
- $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
- $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$
- $(a^3 + b^3) = (a + b)(a^2 - ab + b^2)$
- $(a^3 - b^3) = (a - b)(a^2 + ab + b^2)$

3 Surds and Indices

Rules for powers and roots:

- $a^m \times a^n = a^{m+n}$
- $a^m \div a^n = a^{m-n}$
- $(a^m)^n = a^{mn}$
- $a^0 = 1$
- $a^{-n} = \frac{1}{a^n}$

4 Fraction Simplification

- **Mixed Fraction:** $x\frac{a}{b} = \frac{xb + a}{b}$
- **Reciprocal Division:** $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$

5 Square and Cube Shortcuts

Memorizing these basics significantly increases calculation speed:

Number	Square (x^2)	Cube (x^3)
11	121	1331
12	144	1728
15	225	3375
20	400	8000
25	625	15625

6 Solved Example

Problem: Simplify $[20 + \{15 - (12 \div 3 \times 2)\}]$

Solution:

1. Solve inner bracket $(12 \div 3 \times 2)$:
 $4 \times 2 = 8$
2. Solve curly bracket $\{15 - 8\}$:
 7
3. Final addition $[20 + 7]$:
27